a Composable harness evolution

Current capability map

Plugin	Responsibility
Observer	What is known?
Guide	What comes next?
Control	Which action is valid?

Execution trajectories

- Observation
- Tool call
- Receipt / error
- Next action
- Outcome

2 Improver workspace

Attribute the gap
Evidence + current abilities

Design the mechanism
Trigger · state · handlers

3 Candidate

- skill
- plugin sources
- composition

DSH execution · one task

Fresh episode state

Context guide
guide()

Frozen solver
propose action

Action control
check(call)

Shared state
facts · pending operation

Receipt guidance
explain(call, receipt)

Receipt observer
observe(call, receipt)

Environment
native tool + scorer

4 Evaluate and select

Parent ✓ × ✓ ×

Candidate ✓ ✓ × ✓

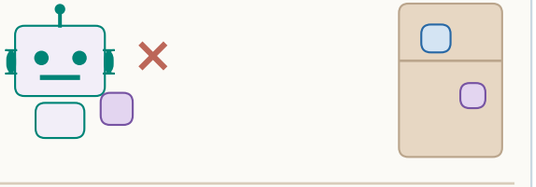
Task scores · hook effects
Repair / regression

Retained code + measured feedback
Next parent harness

Selected harness
Held-out test

b ALFWorld case

1 Observe



Pickup blocked · hand occupied

Pickup fails; the agent loses the pending target

2 Explain the gap

Action checks exist.
Intent is not preserved after failure.

3 Build the repair

Pending-target mechanism

```
observe → save pending target
guide   → free hand; resume
check   → protect next action
observe → clear on success
```

4 Measure the behavior

Recover the interrupted operation
Compare repairs and regressions